Creative Construction: Knowledge Sharing and Cooperation Between Firms*

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Abstract

I provide evidence suggesting that knowledge flows measured by patent citations reflect intentional knowledge transfers rather than involuntary spillovers. Citations to a firm's patents come predominantly from a small set of business partners and are highly firm-specific: inventors who cite a patent at one firm substantially reduce their citations to that patent after moving to another firm. Over time, knowledge transfers became narrower but deeper: firms received citations from a smaller set of firms, while each relationship spanned a broader set of patents. I develop a model that rationalizes these trends through rising technological complexity.

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1 Introduction

Knowledge flows among firms can take two primary forms: involuntary spillovers (Romer, 1990; Aghion and Howitt, 1992) and intentional transfers (Arora, Fosfuri, and Gambardella, 2001; Gans and Stern, 2003). The degree of control firms exert over the use of their knowledge shapes a wide range of economic outcomes, including the private and social returns to R&D (Bloom, Schankerman, and Van Reenen, 2013; Arqué-Castells and Spulber, 2022), geographic location choices (Saxenian, 1996; Shaver and Flyer, 2001; Alcácer and Chung, 2007), and firm boundaries (Teece, 1980; Liebeskind, 1996; Fadeev and Powell, 2026).

In this paper, I evaluate whether knowledge flows measured by patent citations primarily reflect involuntary spillovers or intentional knowledge transfers. The existing literature uses citations in both ways: most studies treat them as proxies for spillovers (e.g., Jaffe, Trajtenberg, and Henderson 1993; Singh and Marx 2013; Akcigit and Kerr 2018; Balsmeier, Fleming, and Lück 2023), while a smaller body of work views them as measures of intentional transfers (e.g., Ziedonis 2004; Galasso and Schankerman 2015).

I provide evidence that knowledge flows measured by citations are likely to reflect intentional transfers rather than involuntary spillovers. First, I show that business partners — such as suppliers, customers, or research collaborators — account for about 73% of inter-firm citations to patents assigned to publicly listed U.S. companies and granted between 2001 and 2014 — the sample in which I can observe business relationships between cited and citing firms.

Table 1: Example of a patent with a high concentration of citations

Patent Number	Assignee	Total Number of Citations	% of Citations from Amkor Technology Inc
5877043	IBM	218 top 0.005%	94%

Second, forward citations — citations received by a patent — are concentrated across firms and became more concentrated over time. For example, the IBM patent in Table 1 is one of the most cited patents granted in the U.S. and yet almost all of its citations come from its input supplier, Amkor Technology. More generally, the top 1% of the most cited patents granted in 2000 received about half of their citations from a single firm, and this share rose to 77% for the most cited patents granted in 2014.

Third, patents citing one firm's patent cite an increasing number of additional patents of that firm. For example, about 70% of the Amkor patents citing the IBM patent in Table 1 also cite a cluster of five additional IBM patents. More generally, for each top-cited patent, I identify citing patents of the firm responsible for the largest share of its citations. For cited patents granted in 1976, a patent of this top-citing firm cited, on average, only 0.2 additional patents of the cited patent's firm. But by 2014, this number had risen to 12.6.

I show that these patterns reflect differences across firms in their *citation rates*. For example, Amkor could account for most citations to the IBM patent simply because it holds more patents than other firms. Instead, Amkor accounts for most citations to the IBM patent because a large *share* of its patents in the citing technological areas cite it, while several firms hold more patents in these areas than Amkor but do not cite it at all. Similarly, the five IBM patents frequently co-cited with the IBM patent are cited far more often than IBM's patent share in the cited technological areas would imply. More generally, business partners cite one another at higher rates than non-partners, top-citing firms cite patents at higher rates than other firms in the same citing technological areas, and patents of the cited firm are cited at higher rates than patents of other firms in the same cited technological areas.

Finally, I show that citation rates are specific to firms rather than inventors. For example, around half of the Amkor inventors who cited the IBM patent later patented at other firms, yet none cited it from those firms. More generally, I compare inventors who move across firms with comparable inventors who remain at the origin firm, restricting attention to the same technological areas before and after the move. Before the move, movers and stayers have similar citation rates, 61.7% and 65.4%, respectively. After the move, these rates fall to 16% for movers and 52.6% for stayers, yielding a difference-in-differences of −33 percentage points.

Given this evidence, I interpret citations as a proxy for intentional knowledge transfers and develop a theory explaining the changes in these transfers over time. The main idea is that rising technological complexity makes knowledge transfers more selective while increasing the amount of knowledge transferred within each relationship.

Consider a firm that owns multiple patents, each embodying a separate piece of codified knowledge. The firm can license these patents to other firms for use in their applications. I refer to these firms as adopters. Applications can be *discrete* or *complex*. Discrete applications use knowledge embodied in one patent, and transferring this knowledge is costless. Complex applications use knowledge from multiple patents and additionally require *tacit* knowledge about how to combine them. Transferring this tacit knowledge is costly — for example, it may require sending the patent owner's engineers to the adopter (Teece, 1977; Arora, 1995). If an adopter implements its application, it files for a patent and cites the patents it uses.

For discrete applications, the firm licenses its patent whenever the application generates positive value. The adopter then cites a *single* patent. For complex applications, licensing occurs only when the application's value exceeds the cost of transferring tacit knowledge, making licensing more selective. Conditional on licensing, the adopter cites *multiple* patents.

I argue that over time the share of complex applications grew relative to discrete applications.

As a result, knowledge transfers became more selective but covered a broader set of patents. More selective transfer increases the concentration of citations across firms, while broader use of the patent owner's knowledge increases the number of its patents cited together. Thus, rising technological complexity can rationalize both trends in citation patterns.

I provide evidence consistent with this mechanism. I use the classification of technologies into discrete, such as pharmaceuticals, and complex, such as semiconductors, from World Intellectual Property Organization (2011). First, I show that the share of corporate patents in complex technologies increased from 55.6% in 1976 to 83.6% in 2014. Second, complex technologies exhibit both more selective citation flows across firms and greater co-citation of multiple patents from the same firm.

Literature. The results in this paper provide an alternative interpretation of some existing evidence based on patent citations. By definition, knowledge spillovers occur regardless of their effect on the original knowledge holder. As a result, the literature that treats citations as a measure of spillovers typically asks why some inventors are better able to absorb knowledge embodied in a patent than others: for instance, they may be geographically closer to the inventors of the cited patent (Jaffe, Trajtenberg, and Henderson, 1993), have prior experience in the relevant field (Balsmeier, Fleming, and Lück, 2026), or have interpersonal connections with the cited patent's inventors (Singh, 2005). A complementary question is why the patent owner has incentives to transfer its knowledge to some firms but not others. This framing does not negate the importance of knowledge absorption but broadens the focus from the characteristics of inventors who use knowledge to the incentives of firms that produce knowledge.

Consider, for example, the geographic localization of citations (Jaffe, Trajtenberg, and Henderson, 1993). This evidence is often interpreted through the lens of inventors: proximity facilitates knowledge diffusion through serendipitous interactions among them (Saxenian, 1996; Atkin, Chen, and Popov, 2022). Alternatively, citations may be localized because business partners co-locate near each other and firms have incentives to transfer knowledge to them. Indeed, one of the leading explanations for industrial agglomeration is that suppliers and customers co-locate to reduce transportation costs (Ellison, Glaeser, and Kerr, 2010).

The paper is also related to the literature on markets for technology (Arora, Fosfuri, and Gambardella, 2001; Gans and Stern, 2003; Ziedonis, 2004; Galasso and Schankerman, 2015; Hegde and Luo, 2017). This literature studies the determinants of intentional knowledge transfers across firms. I contribute by showing that patent citations likely proxy such transfers and that, over time, these transfers became more selective across firms but more intensive within each transfer. I develop and test a model that explains these trends.

Finally, the paper connects to the literature that examines whether patent citations capture knowledge flows. Using a survey of inventors, Jaffe, Trajtenberg, and Fogarty (2000) show that citations are a valid but noisy measure of knowledge flows. They do not distinguish, however, whether these flows are involuntary spillovers or intentional transfers between collaborating firms. Using survey data on French firms, Duguet and MacGarvie (2005) show that firms reporting technology acquisition from a particular country through channels such as R&D cooperation and licensing are more likely to cite patents from that country. I show that most inter-firm citations occur between business partners and document their concentration, clustering, firm-specificity, and evolution over time.

Outline. Section 2 describes the data. Section 3 documents new facts about patent citations. Section 4 develops a model of technological complexity and provides evidence consistent with its predictions. Section 5 concludes.

2 Data and Background

This section describes the data on publicly listed companies, the patent data, and the data on business relationships between firms. All details are provided in Appendix A.

Compustat. I use Compustat North America to identify publicly listed firms, the locations of their headquarters, and their Standard Industrial Classification (SIC) codes. In a robustness exercise, I also use Compustat Global to identify publicly listed firms outside the United States and Canada.

Patents. I use the universe of utility patents granted by the U.S. Patent and Trademark Office (USPTO) between 1976 and 2019. The data come from PatentsView (version from March 2020). A utility patent is granted "for inventing a new or improved and useful process, machine, article of manufacture, or composition of matter."[1]

A patent document includes a written description of the invention, references to prior art (e.g., earlier patents), and claims defining the scope of the intellectual property rights. To be patentable, an invention must be patent-eligible, useful, novel, and non-obvious, and the application must satisfy disclosure requirements (Scotchmer, 2004, ch. 3). Patent examiners use prior art to evaluate whether the claimed invention is novel and non-obvious. Applicants have a "duty of candor" to disclose relevant prior art of which they are aware. Failure to satisfy this duty may render a patent unenforceable. Citations to prior art help delimit the scope of patent claims and therefore serve a legal function in defining the boundaries of intellectual property rights over an invention.

Citations are listed on the front page of a granted patent in the "References Cited" field. When the list is long, the field continues over several pages. Following the literature, I refer to

[1] https://www.uspto.gov/patents/basics/essentials

citations in this field as *front-page citations*. They are the standard citation measure, and I use them throughout the paper. Patents may also cite prior art in the body of the text (Bryan, Ozcan, & Sampat, 2020; Verluise et al., 2026). In a robustness exercise in Section 3.1, I use the data on in-text patent-to-patent citations constructed by Verluise et al. (2026).

Most granted patents contain information about assignees — parties who own a patent. Autor et al. (2020) provide a matching of patent assignees to names of publicly listed firms in Compustat North America and Compustat Global for patents granted between 1976 and 2015. I extend their match through 2019 and for private firms.

Business Partners. I combine three data sets to identify relationships between cited and citing firms. First, the FactSet Revere Supply Chain Relationships data set compiles business relationships disclosed in public sources, including filings with the U.S. Securities and Exchange Commission (SEC), investor presentations, and press releases. The data record each year in which a relationship is reported. Relationship types include supplier–customer links, research collaborations, licensing agreements, joint product offerings, and ownership links such as joint ventures. Second, I use Compustat Segments to identify supplier–customer relationships. Finally, I use USPTO patent reassignment records to identify firms that have traded patents with one another.

For each relationship, I define the start year as the first year in which it is reported. FactSet records an open-ended end date for relationships that are ongoing as of the latest data vintage. I do not assign an end year to these relationships. For all other relationships, I define the end year as the last year in which the relationship is reported. I use relationships reported between 2003 and 2022, the period covered by all three data sets.[2]

Table B1 in Appendix B provides the full list of relationship types I use. I refer to a relationship of any of these types as a *partnership*, and to the two firms as *business partners*.

Descriptive Statistics. Table 2 reports descriptive statistics for the three data sets. Compustat North America contains 35,669 firms headquartered in the United States, of which 30.5% hold at least one utility patent granted between 1976 and 2019.

The patent data contain 6.6 million utility patents. Firms account for 85.4% of these patents: 30.1% are assigned to publicly listed U.S. firms, 27.6% to publicly listed foreign firms, and 27.7% to private firms. The remaining patents are assigned to individuals, universities and research institutes, government bodies, or have no recorded assignee.

The relationship data contain 797,725 partner pairs among publicly listed U.S. firms. Partnerships are rare: only 0.13% of all possible pairs of U.S. public firms have a recorded

[2]FactSet provides the broadest coverage of relationship types, but its coverage begins in 2003. Compustat Segments and USPTO reassignment records provide coverage beginning in 1976 and 1968, respectively. Relationships whose reports end before 2003 are excluded.

Table 2: Descriptive Statistics

Panel A. Publicly listed U.S. firms	
Publicly listed U.S. firms	35,669
Share with at least one granted patent, 1976–2019	30.5%
Panel B. Utility patents granted 1976–2019	
Utility patents	6,559,305
Share assigned to publicly listed U.S. firms	30.1%
Share assigned to publicly listed foreign firms	27.6%
Share assigned to private companies	27.7%
Share with other or missing assignees	14.7%
Panel C. Business partnerships between U.S. publicly listed firms	
Partner pairs	797,725
Share of records with no end year	8.4%
Median length of closed records (years)	3
Partner pairs as share of all U.S. public firm pairs	0.13%
Share of firms with at least one partner	23.8%
Share of partner links held by top 1% most-connected firms	10.5%

Notes: Panel A shows the number of firms recorded in Compustat North America with headquarters in the United States (coverage 1950–2024). Panel B shows the distribution of patents by assignee types. A patent with several assignees is classified into a single category by the following priority: if any assignee is a publicly listed U.S. firm, the patent is classified as publicly listed U.S.; otherwise, if any assignee is a publicly listed foreign firm, as publicly listed foreign; otherwise, if any assignee is a private company, as private. Patents whose assignees are all individuals, universities and research institutes, or government bodies, or whose assignee is not recorded, are classified as other or missing. Publicly listed firms are identified in Compustat North America and Compustat Global. In Panel C, partner pairs are unordered pairs of publicly listed U.S. firms with at least one partnership reported between 2003 and 2022. "No end year" means that the source reports no end date. Lengths are inclusive, so a relationship reported within a single calendar year has length one.

partnership. About 23.8% of firms have at least one partner, and the top 1% of the most connected firms account for 10.5% of all partner links. The median duration of partnerships is three years. The reported duration likely understates the true duration because relationship dates are self-reported by firms.

Table B1 also reports the share of partner pairs with each type of partnership. Supplier–customer links are the most common, appearing in 83.1% of partner pairs, followed by research collaborations, which appear in 29.7%.

3 Facts About Patent Citations

This section documents several new facts about patent citations: whether cited and citing firms are business partners, the concentration of citations across firms, the clustering of citations across patents, and the citation patterns after inventors move between firms.

For each fact, I compute a statistic from the observed citation network and compare it with the distribution the statistic would have if citations were randomly assigned under a specific null hypothesis. These tests are designed to separate citation behavior from other features of the data that can mechanically affect citation-based statistics.

3.1 Business Partners

This section analyzes whether cited and citing firms are business partners.

Sample. The unit of observation is a citation: an ordered pair of a cited and a citing patent, in which the citing patent lists the cited patent as prior art. I focus on citations satisfying four conditions: (i) each patent is assigned to a single publicly listed U.S. firm; (ii) the cited patent is granted between 2001 and 2014; (iii) the two patents are assigned to different firms, that is, self-citations are excluded; (iv) the citing patent is granted no earlier than the cited patent and within five years of it.

Conditions (i) and (ii) reflect the coverage of the data on business relationships. The data are based on information disclosed in public sources and therefore primarily cover reporting by publicly listed firms. The main sample focuses on firms headquartered in the United States, and I consider publicly listed foreign firms in a robustness exercise. The lower bound in condition (ii) reflects the availability of business-relationship data,[3] while the upper bound ensures that each cited patent has a complete five-year citation window before the patent data end in 2019. Condition (iii) follows from my focus on inter-firm knowledge flows. Condition (iv) ensures that cited patents granted in different years have the same five-year window to accumulate citations. I consider a ten-year window in a robustness exercise.

Main Statistic. The main statistic in this section is the share of citations in which the cited and citing firms are business partners. I refer to this statistic as the *partner share*.

The partner share depends on both the prevalence of partners among the firms that could have made the citations and the rate at which partners cite relative to non-partners. For example, among the citing patents assigned to publicly listed U.S. firms, every firm citing the

[3] I include cited patents granted in 2001 and 2002 because the start of FactSet coverage censors when relationships are first observed: a relationship active in 2001–2002 that continues into 2003 enters the data, with 2003 as its first reported year. In a robustness exercise, I restrict the sample to cited patents granted from 2003 to 2014.

IBM patent in Table 1 is a business partner of IBM. This high share may reflect the fact that IBM has many partners — indeed, IBM is among the top 1% of firms by number of partners — or that IBM's partners cite its patents at a higher rate than non-partners. To separate these two factors, I compare the actual partner share with the following random benchmark.

Random Benchmark. For each citation, I randomly assign the citing patent to a publicly listed U.S. firm drawn uniformly from firms in the same four-digit SIC industry as the actual citing firm. Candidate firms must be covered by Compustat during the years of the pair's citing activity. I then recompute the partner share. I repeat the procedure 1,000 times and report the mean partner share across draws, together with the 2.5th–97.5th percentile range.

The intuition behind the benchmark is as follows. I treat firms operating in the same industry and period as the actual citing firm as firms that could plausibly have made the citation. Within each such group, the randomization gives every firm the same probability of making the citation, while holding fixed the network of business partnerships. The resulting partner share therefore reflects only the prevalence of the cited firm's partners among the candidate firms in the industries from which its citations originate. The positive gap between the actual partner share and the benchmark reflects the higher rate at which partners cite relative to non-partners in the same industries.

Example 1. To illustrate the construction of the random benchmark, consider the IBM patent in Table 1. Among its citations, 208 come from citing patents assigned to publicly listed U.S. firms.[4] The citing firms are Amkor (204 citations), Eli Lilly (2), Honeywell (1), and Medtronic (1), and all four are business partners of IBM in the relationship data, so the actual partner share is 100%. Under the random benchmark, each citation is reassigned within the citing firm's industry. Amkor's industry contains 250 candidate firms, of which 113 are IBM's partners, so a citation reassigned in this industry lands on a partner with probability $113/250 \approx 45\%$. Across all 1,000 draws, the benchmark partner share is 44.5%, with a 2.5th–97.5th percentile range of $[37.5, 51.0]$, against the actual 100%. Even for a firm in the top 1% of partner connections, citations concentrate on partners far beyond what their prevalence produces.

Main Result: Partner Share. Figure 1 shows that 73.1% of citations occur between business partners. Under the random benchmark, this share is 21.0% on average, with a 2.5th–97.5th percentile range of $[20.9, 21.1]$. Thus, business partners cite each other 3.5 times as often as they would if citations were assigned at random within the citing firms' industries

Composition of Partners. Figure B1 in Appendix B decomposes citations between partners by the type of relationship. Supplier–customer relationships account for the largest share,

[4] As in Table 1, I consider all citations to the patent. Of the 218 citations, two citing patents have multiple assignees, three are IBM's own patents, and five are assigned to foreign or private firms.

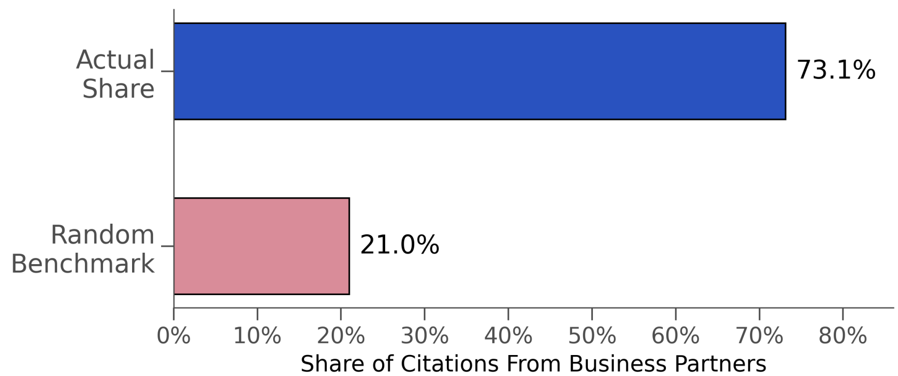


Figure 1: Share of citations from business partners

This figure shows the share of patent pairs in which the cited and citing firms are business partners (the "Actual Share" bar). The "Random Benchmark" bar shows the mean partner share across 1,000 draws in which citations are randomly reassigned among publicly listed U.S. firms in the same four-digit SIC industry as the actual citing firm. The cited firm is excluded from the set of possible citing firms. The 2.5th–97.5th percentile range across draws is $[20.9, 21.1]$.

41.9%, followed by research collaborations, 28.4%. This distribution should be interpreted with caution. Relationship types are self-reported, and firms may have stronger incentives to disclose some types of relationships than others. The correct interpretation of Figure B1 is that partners responsible for most citations had *at least* a supplier–customer relationship or a research collaboration. Many of these partners likely had additional relationships, including licensing agreements, that are not separately reported.

Timing. Figure B2 shows that among the partner citations, 60.6% have citing-patent application years within the reported window of the partnership, 33.9% before it, and 5.5% after it. Since the reported window lies within the actual period of the partnership, 60.6% is a lower bound on the share of partner citations that occur during the partnership.

Robustness. Table B2 in Appendix B shows the robustness of the main result. Verluise et al. (2026) suggest that in-text citations — references made in the body of the patent text — better reflect inventors' knowledge than front-page citations, which are influenced by legal practices and negotiations with the patent examiner. Among in-text citations, 74.8% occur between partners, against a random benchmark of 18.9% — in line with the main result.

The result is also unlikely to reflect strategic withholding of citations. Lampe (2012) argues that applicants withhold relevant prior art to increase the scope of their claims. Applied to my setting, this argument suggests that the partner share could be high if applicants withhold citations to non-partners but are less willing to do so with business partners — for example, because partners directly negotiate over the boundaries of intellectual property rights. To address this concern, I consider examiner–added citations. Examiners search prior art independently and have no incentives to withhold it. The partner share among examiner-added citations is 77.9%, against a random benchmark of 23.2% — as high as in the full sample. In addition, Kuhn, Younge, and Marco (2023) reassess Lampe's analysis and find no evidence that applicants withhold citations.

The result is also robust to the length of the citation window. With a ten-year window (and cited patents restricted to grants through 2009, so that the full window lies within the patent data), 70.1% of citations occur between partners, against a benchmark of 20.0%. Restricting cited patents to grants from 2003 onward, so that the citation window lies within the coverage of the FactSet data, yields a partner share of 73.4%, against a benchmark of 21.4%.

Finally, the result is not specific to U.S. firms. Among publicly listed foreign firms, 71.2% of citations occur between partners, against a benchmark of 5.8%. This benchmark is lower than in the U.S. sample because *reported* partnerships among foreign firms are sparser in the data.

3.2 Concentration of Citations

This section documents the concentration of citations across firms.

Sample. The sample consists of citations that satisfy the following conditions: (i) the cited and the citing patent are each assigned to a single firm; (ii) the cited patent is granted between 1976 and 2014; (iii) the cited and citing firms are different; (iv) the citing patent is granted no earlier than the cited patent and within five years of it; (v) the cited patent is among the top 1% of patents by the number of citations received within its grant year and technological class — group level in the Cooperative Patent Classification (CPC).

The main differences relative to the sample in Section 3.1 are that the sample includes private and foreign firms and that cited patents go back to 1976. The main statistic below does not depend on the partnership data and therefore allows broader coverage of patents. Condition (v) focuses on the most-cited patents, which the literature typically treats as the most valuable (Trajtenberg, 1990; Hall, Jaffe, and Trajtenberg, 2005). The number of citations received is computed from the citations satisfying conditions (i)–(iv). The sample contains 47,829 cited patents. I provide robustness checks for alternative samples below.

Main Statistic. For each cited patent k, let n_{ik} denote the number of citations made by patents assigned to firm i. I define the *concentration* of citations as the largest share of patent k's citations accounted for by a single citing firm, $C_k = \max_i\{n_{ik}/\sum_j n_{jk}\}$. I report the average concentration across cited patents in each grant year.

The number of citations n_{ik} depends on both the number of firm i's patents in the relevant technological areas and its *citation rate* — the share of these patents that cite patent k. Citation concentration can therefore reflect differences across firms in either patent counts or citation rates. For example, Amkor may account for 94% of citations to the IBM patent in Table 1 because it holds a large share of the patents in the technological areas that rely on the IBM patent, or because its patents cite the IBM patent at a higher rate than those of other firms. To separate these two factors, I compare the actual concentration of citations with the following random benchmarks.

Random Benchmarks. Consider a cited patent k. I divide the patents citing k into disjoint groups based on application year and narrow technological class — the main-subgroup level of the CPC.[5] For each group, I identify all patents, both citing and non-citing, with the same application year and narrow technological class that satisfy conditions (i), (iii), and (iv) with respect to patent k. I treat the non-citing patents in these groups as patents that could plausibly have cited patent k but did not.[6] In a robustness exercise below, I refine this comparison by also matching patents on textual similarity.

I consider two benchmarks. Under the first, with equal patent counts, I assign each of the group's citations independently to one of the firms with a patent in the group, each firm with equal probability. Under the second, which preserves patent counts, I randomly permute the observed citations among the group's patents and recompute the concentration of citations to patent k. I repeat each procedure 1,000 times. In each draw, I compute the average concentration across cited patents in each grant year, and I report the mean of this average across draws, together with its 2.5th–97.5th percentile range.

Together, the two benchmarks decompose the concentration of citations into three sources. The first benchmark shows the baseline concentration driven by the finite number of potential citing firms: for example, if a patent receives all of its citations from a single group with only two potential citing firms, its concentration cannot be lower than 50%. The second benchmark adds the differences in firms' patent counts: every patent within a group has the same probability of citing patent k, so firms receive citations in proportion to their patent counts. The difference between the two benchmarks is therefore the *excess concentration* from the dispersion of patent counts across the firms with at least one patent in the group. The remaining gap — between the actual concentration and the second benchmark — isolates the *excess concentration* from differences in citation rates across firms: firms citing patent k more intensively than their patent counts imply.

Example 2. To illustrate the construction of the random benchmarks, consider the IBM patent

[5] The main-subgroup CPC lists 7,137 detailed technological classes. This classification is substantially more detailed than the ones typically used in the literature. For a discussion of the granularity of technological classes in citation analysis, see Thompson and Fox-Kean (2005) and Henderson, Jaffe, and Trajtenberg (2005).

[6] In Section 3.1, the analysis is focused on *firms* that could have potentially cited a particular firm, as the data on relationships are defined at the firm-pair level. In this section, the analysis is focused on *patents* that could have potentially cited a particular patent.

in Table 1. The patents citing it are divided into 68 disjoint groups based on application year and narrow technological class.[7] The largest group consists of 16 citing patents, all assigned to Amkor, in class H01L23 with application year 2003. There are 1,321 patents in this group, assigned to 295 firms, of which only 19 are assigned to Amkor.[8]

Under the first benchmark, Amkor would account for $16 \cdot \frac{1}{295} \approx 0.05$ of the group's citations on average. Under the second benchmark, Amkor would account for $16 \cdot \frac{19}{1{,}321} \approx 0.2$ of the group's citations. Sixteen firms have more patents than Amkor in this class and year. For example, Intel and Micron Technology have 122 and 101 patents, respectively, yet none of their patents cite IBM's patent, while 16 of Amkor's 19 patents do. Within this group, Amkor's citation rate is thus $\frac{16}{19}$, while every other firm's rate is zero. Repeating the randomizations across all 68 groups produces a mean concentration of 1.7%, with a 2.5th–97.5th percentile range of $[1.4, 2.3]$, under the first benchmark, and 6%, with a range of $[4.1, 8.3]$, under the second, compared with the actual concentration of 94%. Thus, Amkor accounts for most citations to the IBM patent because its patents cite it at a higher rate than other companies — not because Amkor holds more patents in the relevant technological areas, and not because few firms are active in them.

Main Results. Figure 2 shows the average concentration of citations by cited grant year, together with the two benchmarks. For patents granted in 1976, the actual concentration is 52.2%, compared with 37.7% under the benchmark with equal patent counts and 39.5% under the benchmark that preserves patent counts. By 2014, the actual concentration has risen to 77.2%, while the benchmarks have fallen to 14.5% and 20.3%.[9]

The benchmarks decompose the concentration into the three sources annotated in Figure 2. In 1976, most of the concentration ($37.7/52.2 \approx 72\%$) is the baseline concentration due to the finite number of potential citing firms. Another 3% is the excess concentration due to differences in patent counts — the gap between the two benchmarks — and the remaining 24% is the excess concentration due to differences in citation rates. By 2014, the composition has reversed: the finite number of firms accounts for 19% of the concentration, differences in patent counts for 8%, and differences in citation rates for 74%. Over these four decades, patenting within narrowly defined technologies spread across substantially more firms, pushing both benchmarks down — and yet the actual concentration rose.

The large gap between the actual concentration and the benchmark that preserves patent

[7] To keep the example consistent with the numbers in Table 1, I consider all citations to IBM's patent.

[8] Amkor has one additional patent in this class and year that is assigned jointly to Amkor and another firm. Among the 218 citations, three citing patents are assigned to IBM and therefore constitute self-citations, two have multiple assignees, and one has no recorded assignee. I keep these six citations fixed in the randomizations.

[9] Benchmark lines report means across 1,000 draws. The yearly averages vary little across draws: the widest 2.5th–97.5th percentile ranges both occur in 1980 and extend from 36.6% to 37.6% for the benchmark with equal patent counts and from 38.5% to 39.5% for the benchmark that preserves patent counts. I therefore omit the ranges from the figure.

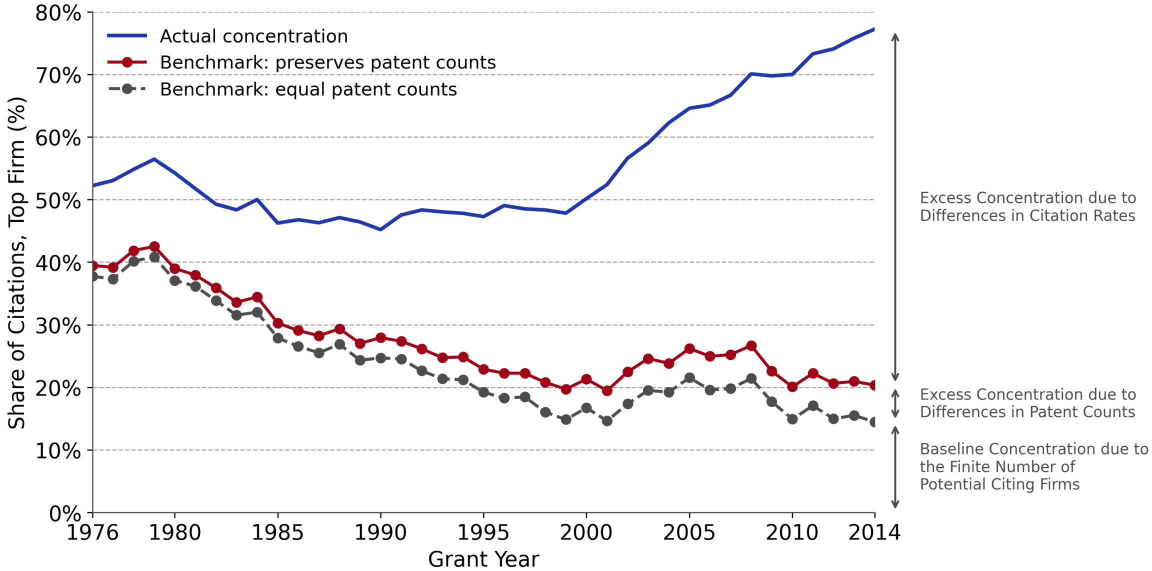


Figure 2: Concentration of Citations

The figure shows, by cited grant year, the average concentration of citations — the largest share of a cited patent's citations accounted for by a single citing firm. The sample includes the top 1% of the most cited patents within each grant year and technological class. Under the benchmark that preserves patent counts, citations originating from each group defined by application year and narrow technological class are randomly permuted among the group's patents, so that, in expectation, firms receive citations in proportion to the number of patents they hold in each group. Under the benchmark with equal patent counts, each citation from a group is equally likely to come from any firm with a patent in the group, regardless of the firms' patent counts. Benchmark lines report means across 1,000 draws. The annotations at the right decompose the 2014 concentration into the part produced by the finite number of potential citing firms (the level of the equal-counts benchmark), the part added by differences in patent counts across firms (the gap between the two benchmarks), and the remainder, reflecting differences in citation rates across firms.

counts shows that technological specialization alone cannot explain citation concentration. If firms were fully specialized — so that all patents within each group belonged to a single firm, although that firm could differ across groups — the benchmark would equal the actual concentration. The gap between the actual concentration and this benchmark therefore reflects differences in citation rates across firms even within narrowly defined technologies.

Robustness. I do four types of robustness exercises, summarized in Figure B3 in Appendix B. First, I recompute the concentration measure on different samples. Panel (a) shows the concentration over time for the top 5% of the most-cited patents, and panel (b) for the sample of patents assigned to publicly listed U.S. firms, as in Section 3.1.

Second, in panel (c), I measure the concentration of citations by the Herfindahl index of citation shares across citing firms instead of the top-firm share.

Third, in panel (d), I take a more stringent approach to identifying the non-citing patents that could have cited a particular patent: in addition to the same application year and narrow technological class, I require textual similarity to the citing patents. I compute an embedding of each patent's title and abstract with the PaECTER model (Ghosh et al., 2024), which is trained on patent examiners' citations, so that patents linked by a citation are close in the embedding space. For each cited patent, I measure the similarity among its citing patents, and I keep a non-citing patent in the group only if it exhibits at least as much similarity to the citing patents as the citing ones do among themselves. I then recompute the random benchmark within the trimmed groups.

Finally, in panel (e), I control for repeated citations across continuation patents, which can cause a single invention to count as several citing patents. I collapse citing patents to patent families: each family counts as a single citing patent, represented by its earliest member, whose application year, technological class, and assignee are used throughout; a family cites the focal patent if any of its members does. The candidate pool and the selection of the most-cited patents use the same family-collapsed counts.

All four exercises show a pattern similar to that in Figure 2: differences in citation rates account for relatively little of citation concentration early in the sample, but account for most of it by 2014.

3.3 Clustering of Citations

This section examines whether the firm responsible for most citations to a highly-cited patent also tends to cite other patents assigned to the patent's firm.

Sample. I use the same sample as in Section 3.2.

Main Statistic. Consider a cited patent k, assigned to firm A. I refer to a firm with the largest number of patents citing k as a *top-citing firm*. For each top-citing firm i, let n_{ik} denote the number of its patents that cite k. For each such citing patent c, let m_{cA}^{-k} denote the number of patents other than k that are assigned to firm A and cited by patent c. I define the *citation clustering* around patent k and top-citing firm i as

$$P_{ik} = \frac{1}{n_{ik}} \sum_{c \in \mathcal{C}_{ik}} m_{cA}^{-k},$$

where $\mathcal{C}_{ik}$ is the set of patents assigned to firm i that cite patent k. Thus, P_{ik} is the average number of other patents assigned to the cited firm that are cited by a patent of its top-citing firm. If several firms are tied as top-citing firms, I average P_{ik} across them to obtain P_k. I report the average of P_k across cited patents in each grant year.

The number m_{cA}^{-k} depends on the number of backward citations made by patent c, the share of patents assigned to firm A in the technological areas reached by these citations, and the rate at which c cites A's patents relative to patents of other firms. For example, an Amkor patent may cite many other IBM patents because it makes many backward citations and IBM holds a large share of patents in the technological areas it cites, or because it cites IBM patents at a higher rate than patents of other firms in the same technologies. To separate these factors, I compare the actual clustering of citations with the following random benchmarks.

Random Benchmarks. Consider a cited patent k, assigned to firm A, a top-citing firm i, and a patent $c \in \mathcal{C}_{ik}$. I divide the backward citations made by c, excluding its citation to k, into disjoint groups based on the grant year and narrow technological class — the main-subgroup level of the CPC — of the cited patents. For each group, I identify all patents with the same grant year and narrow technological class that could have received these citations.[10] I exclude patents assigned to firm i and patent k itself.

I consider two benchmarks. Under the first, with equal patent counts, I assign each citation in a group to one of the firms with an eligible patent in the group, with equal probability across firms. Under the second, which preserves patent counts, I randomly assign the observed citations among the eligible patents in the group, without replacement. Thus, under the second benchmark, firms are more likely to receive citations when they hold more patents in the group, while every eligible patent has the same probability of being cited.

I repeat each procedure 1,000 times and recompute P_k in every draw. For each draw, I compute the average citation clustering across cited patents in each grant year. I report the mean across draws, together with its 2.5th–97.5th percentile range.

Together, the two benchmarks decompose citation clustering into three sources. The first benchmark captures the baseline clustering generated by the finite number of potential cited firms. The gap between the second and the first benchmarks captures the additional clustering due to differences in patent counts. The remaining gap between the actual statistic and the second benchmark reflects differences across cited firms in the rates at which their patents are cited by patents of the top-citing firm.

Example 3. To illustrate the construction of the main statistic and random benchmarks, consider again the IBM patent in Table 1. Amkor is its top-citing firm: 204 Amkor patents cite the focal IBM patent.[11] These patents also cite 54 other IBM patents. Table 3 shows the five

[10] The construction parallels the random benchmarks for the concentration measure, except that I use grant year rather than application year. In the concentration exercise, I compare potential *citing* patents with the actual citing patents, so I use application year to match patents created around the same time. Here, I compare potential *cited* patents with the actual cited patents, so I use grant year to match patents that became publicly available as granted patents around the same time. Grant year need not coincide with the first public disclosure of an invention: in particular, since 2001, U.S. patent applications are generally published before grant. It nevertheless provides a consistently observed measure of the timing of granted patents throughout the sample.

[11] As in Example 2, to keep the example consistent with the numbers in Table 1, I consider all citations to IBM's patent.

Table 3: IBM Patents Co-Cited with Patent 5877043 by Amkor

IBM patent	Grant year	CPC class	Number of Amkor patents citing it	Share of 204 Amkor patents (%)
5435057	1995	H05K3	182	89.2
5814877	1998	H01L23	167	81.9
5543657	1996	H01L23	164	80.4
4862245	1989	H01L24	163	79.9
5168368	1992	H01L24	160	78.4
5769989	1998	H01L24	43	21.1
6258192	2001	B32B37	43	21.1
...	...	...	...	...

The table shows the seven other IBM patents cited by the largest numbers of the 204 Amkor patents that cite IBM patent 5877043. The first five patents account for 836 of the 1,176 citation links from these Amkor patents to other IBM patents.

most frequently cited ones, together with the next two most-cited patents. The top five form a tight cluster: 140 of the 204 Amkor patents, or 68.6%, cite all five, and 162, or 79.4%, cite at least four. The sixth-most commonly cited IBM patent, by contrast, is cited by only 43 Amkor patents. Across all 54 other IBM patents, there are 1,176 Amkor–IBM citation links, so the clustering statistic for the IBM–Amkor pair is $\frac{1{,}176}{204} = 5.76$. Thus, an Amkor patent that cites the focal IBM patent cites, on average, 5.76 other IBM patents. The top five account for 836, or 71.1%, of these 1,176 links.

To illustrate the random benchmarks, consider IBM patent 5435057, the first patent in Table 3. It was granted in 1995 and belongs to narrow technological class H05K3. Of the 204 Amkor patents, 182 cite this patent. Altogether, the Amkor patents make 188 citations to patents granted in 1995 in class H05K3. Of these, 186 are to IBM patents: 182 to patent 5435057 and four to another IBM patent, 5404044. The remaining two citations are to a patent assigned to another firm.

There are 232 eligible patents in this grant-year and technological-class group, assigned to 108 firms, of which 27 are assigned to IBM. Under the benchmark with equal patent counts, IBM would therefore receive $188 \cdot \frac{1}{108} \approx 1.7$ citations on average. Under the benchmark that preserves patent counts, IBM would receive $188 \cdot \frac{27}{232} \approx 21.9$ citations on average. In the data, IBM receives 186 of the 188 citations. Thus, within this group, Amkor's citations cluster on IBM patents far more strongly than either the finite number of potential cited firms or IBM's patent count would imply.

Main Results. Figure 3 shows the average citation clustering by cited grant year, together with the two benchmarks. For patents granted in 1976, a patent of the top-citing firm cites, on average, 0.20 other patents assigned to the cited firm, compared with 0.05 under the benchmark

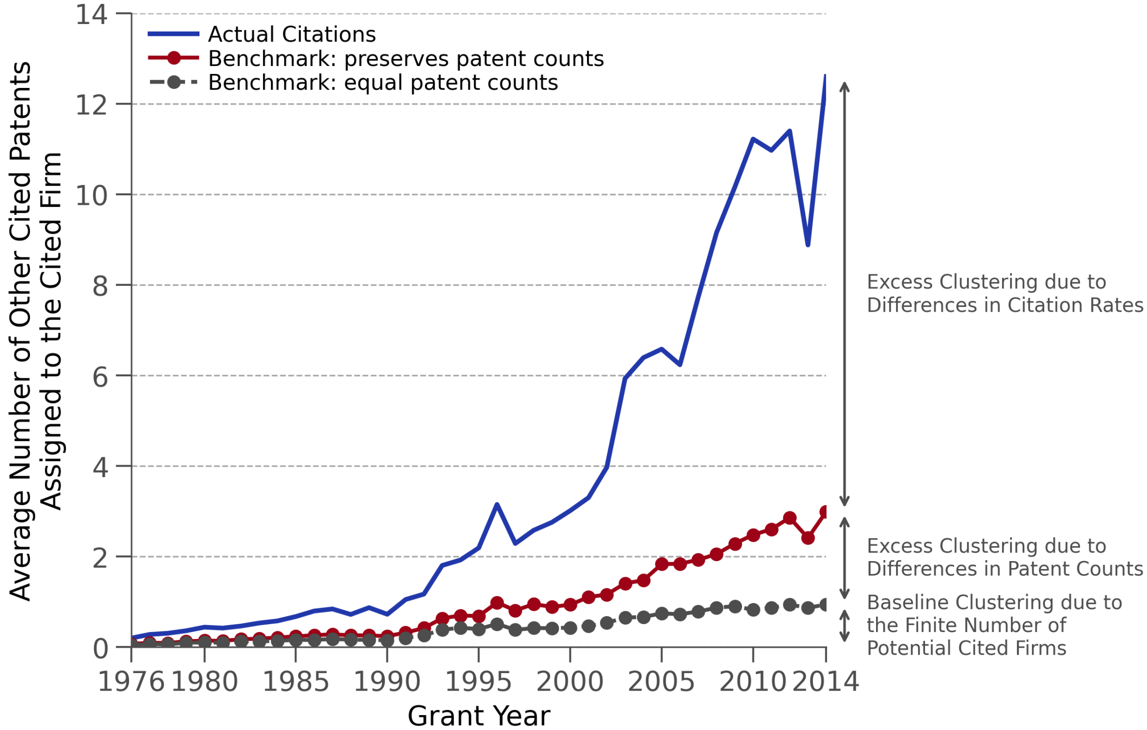


Figure 3: Clustering of Citations

The figure shows, by cited grant year, the average citation clustering—the average number of other patents assigned to the cited firm that are cited by a patent of its top-citing firm. If several firms are tied as top-citing firms, I first average the statistic across the tied firms. Under the benchmark that preserves patent counts, the backward citations made by each citing patent within a group defined by grant year and narrow technological class are randomly assigned among the group's eligible patents. Under the benchmark with equal patent counts, each citation is instead equally likely to go to any firm with an eligible patent in the group. Benchmark lines report means across 1,000 draws. The annotations at the right decompose the 2014 clustering into the part produced by the finite number of potential cited firms, the part added by differences in patent counts across firms, and the remainder, reflecting the tendency of the top-citing firm to cite other patents of the cited firm more often than comparable patents of other firms.

with equal patent counts and 0.07 under the benchmark that preserves patent counts. By 2014, the actual clustering has risen to 12.60 other patents, compared with 0.94 and 2.99, respectively, under the two benchmarks.[12]

The benchmarks decompose citation clustering into the three sources annotated in Figure 3. In 1976, the finite number of potential cited firms accounts for about 23% of the actual clustering (0.05/0.20), differences in patent counts account for another 12%, and differences in citation rates account for the remaining 65%. By 2014, the finite number of firms accounts for only 7%

[12]Benchmark lines report means across 1,000 draws. The yearly averages vary little across draws. The widest 2.5th–97.5th percentile ranges for both benchmarks occur in 2005 and extend from 1.81 to 1.85 under the benchmark that preserves patent counts and from 0.73 to 0.76 under the benchmark with equal patent counts. I therefore omit the ranges from the figure.

of the clustering, differences in patent counts for 16%, and differences in citation rates for 76%. Thus, differences in citation rates account for most citation clustering throughout the sample and become even more important over time.

The concentration and clustering results together point to *firm-pair-specific* knowledge flows. The concentration results in Section 3.2 show that a focal firm's patent is disproportionately cited by one particular firm: the top-citing firm accounts for a larger share of citations than its share of potentially citing patents in the relevant technological areas would predict. The clustering results shows the mirror image: among the backward citations made by patents of the top-citing firm, the focal firm's patents are cited more frequently than their share of potentially cited patents in the relevant technological areas would predict. Thus, the same pair of firms is unusually strongly connected in both directions of the citation exercise.

Robustness. I do three types of robustness exercises, summarized in Figure B4 in Appendix B. First, I recompute the clustering measure on alternative samples. Panel (a) uses the top 5% of the most-cited patents instead of the top 1%, and panel (b) restricts the sample to patents assigned to publicly listed U.S. firms, as in Section 3.1.

Second, in panel (c), I address the concern that citations to several continuation-related patents may mechanically inflate the number of other patents of the cited firm that are cited. I collapse the patents receiving backward citations to patent families and count the number of distinct families rather than distinct patents. Each family is represented by its earliest member, and the randomization pool is collapsed to families in the same way.

Finally, in panel (d), I impose a more stringent requirement on the patents that could have received a backward citation. In addition to having the same grant year and narrow technological class as an actually cited patent, a non-cited patent must be textually similar to the patents actually cited by the same citing patent. I measure textual similarity using PaECTER embeddings (Ghosh et al., 2024). For each citing patent, I measure the similarity among its actually cited patents and retain a non-cited patent only if its similarity to at least one actually cited patent is at least as high as the minimum similarity among the actually cited patents themselves. I then recompute both random benchmarks within these trimmed sets of potential cited patents.

All robustness exercises show a pattern similar to that in Figure 3. Differences in citation rates across cited firms account for the majority of citation clustering, and their importance increases from the 1980s to the 2010s.

3.4 Firm-Specificity of Citations: Evidence From Inventors-Movers

This section separates the roles of firms and inventors in citation patterns by analyzing citations made by inventors who patent at more than one firm.

Sample. I restrict attention to patents assigned to a single firm and exclude firm self-citations. Consider the first application year in which inventor i has a patent citing focal patent q. I restrict the sample to cases in which all of inventor i's patents citing q in this first year are assigned to the same firm A. I then consider each other firm $B \neq A$ at which inventor i subsequently begins to patent, and let m denote the first application year in which i has a patent assigned to B. I refer to inventor i as the *mover*, to firm A as the *origin firm*, to firm B as the *destination firm*, and to year m as the *move year*.

For each (q, i, B), I define the set of *relevant technological classes* as the main-subgroup CPC classes in which inventor i has a patent assigned to A that cites q and has an application year strictly below m. I also identify potential *stayers* j. A stayer is an inventor whose patents citing q in the first application year in which j cites q are all assigned to the same origin firm A, and who has no patents assigned to a firm other than A from year m onward.

The unit of observation is a cited-patent–mover–destination-firm triple, (q, i, B). The sample consists of triples satisfying the following conditions: (i) focal patent q is granted between 1976 and 2014; (ii) before year m, inventor i has at least five distinct patents assigned to A that cite q; (iii) destination firm B does not own patent q; (iv) $m \leqslant 2010$; (v) inventor i has at least one patent assigned to A in the relevant technological classes during $[m-5, m)$ and at least one patent assigned to B in the same classes during $[m, m+5)$; and (vi) there exists a stayer j who has never appeared as a co-inventor on a patent with mover i, who has at least five distinct patents assigned to A that cite q before year m, and who has at least one patent assigned to A in the relevant technological classes in each of the two five-year windows, $[m-5, m)$ and $[m, m+5)$.

Restricting the sample to patents granted by 2014 in condition (i) allows focal patents to accumulate at least five years of citations before the end of the citation data in 2019. Condition (ii) ensures that inventor i had repeatedly relied on patent q before the move and was therefore likely aware of the knowledge embodied in it. Condition (iv) ensures that patents filed at the destination firm during $[m, m+5)$ have application years no later than 2014 and therefore have up to five years to be granted before the patent data end in 2019. Condition (v) ensures that the mover is observed patenting within the same pre-move narrow technological classes both before and after the move. Finally, condition (vi) ensures that each mover can be compared with at least one stayer who had also repeatedly cited patent q before the move and is observed patenting in the same narrow technological classes in both five-year windows. I exclude inventors who have ever appeared as co-inventors with the mover because the mover's transition to the destination firm could itself affect the citation behavior of former collaborators. Restricting the comparison group to inventors who have not patented jointly with the mover reduces this concern. I discuss robustness to alternative sample constructions below.

Main Statistic. For each cited-patent–mover–destination-firm triple $s = (q, i, B)$, I measure the *citation rate* to q before and after the move. For the mover, the pre-move citation rate is the share of inventor i's patents assigned to the origin firm A, filed in the relevant technological classes during $[m-5, m)$, that cite q. The post-move citation rate is the share of inventor i's patents assigned to the destination firm B, filed in the same technological classes during $[m, m+5)$, that cite q. Let $Y^M_{s,\text{pre}}$ and $Y^M_{s,\text{post}}$ denote these two citation rates. For each eligible stayer j in triple s, I calculate the same two citation rates, $Y^j_{s,\text{pre}}$ and $Y^j_{s,\text{post}}$. I use the stayer's patents assigned to the origin firm A in the same narrow technological classes and five-year windows as for the mover. I then average these citation rates across all eligible stayers within s:

$$Y^S_{s,t} = \frac{1}{|J_s|} \sum_{j \in J_s} Y^j_{s,t}, \qquad t \in \{\text{pre}, \text{post}\},$$

where J_s denotes the set of eligible stayers for triple s.

I define the unit-level difference-in-differences as

$$D_s = \left(Y^M_{s,\text{post}} - Y^M_{s,\text{pre}}\right) - \left(Y^S_{s,\text{post}} - Y^S_{s,\text{pre}}\right).$$

The main statistic is the average of D_s across all cited-patent–mover–destination-firm triples:

$$\text{DiD} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} D_s,$$

where $\mathcal{S}$ denotes the final sample of triples.

The statistic compares changes in citation rates between movers and stayers. For both groups, I consider the technological classes in which the mover cited q before the move, so changes in citations are not driven by shifts across technologies.

To assess whether the observed change in citation behavior is unusual relative to the changes among comparable inventors, I construct a within-triple random benchmark.

Random Benchmark. For each triple $s = (q, i, B)$, define the before–after change in the mover's citation rate as $\Delta Y^M_s = Y^M_{s,\text{post}} - Y^M_{s,\text{pre}}$, and define analogously the change for each eligible stayer $j \in J_s$ as $\Delta Y^j_s = Y^j_{s,\text{post}} - Y^j_{s,\text{pre}}$. In each randomization draw, I randomly select one inventor from the mover and eligible stayers in each triple to serve as the pseudo-mover, with the remaining inventors serving as pseudo-stayers. I repeat this procedure 1,000 times, recomputing the DiD statistic in each draw. The random benchmark is centered at zero by construction. I therefore report the 2.5th and 97.5th percentiles of the resulting distribution, together with a finite-randomization left-tail p-value.

Inference. I also report standard errors and confidence intervals for the main statistic. Since the same inventor can appear in multiple triples, either as a mover or as a stayer, different triples need not be independent. I therefore construct a mover–stayer inventor network, in which each inventor is a node and an edge connects each mover to every stayer used as a comparison inventor in one of the mover's triples. I define clusters as the connected components of this network and conduct a cluster bootstrap at the network-component level. In each of 1,000 bootstrap draws, I sample components with replacement and recompute the main statistic. I report the standard deviation of the bootstrap estimates as the standard error and the 2.5th and 97.5th percentiles as the 95% confidence interval.

Example 4. To illustrate the construction of the main statistic and the random benchmark, consider an inventor listed on Amkor patent 6,448,633, filed in 1999, which cites the IBM patent in Table 1. This application year is the first year in which the inventor cites the IBM patent, so Amkor is the inventor's origin firm. The inventor subsequently begins patenting at another firm in 2004. I refer to this inventor as the mover. Before 2004, the mover has six Amkor patents citing the IBM patent, in CPC classes H01L23 and H01L24. These two classes therefore constitute the relevant technological classes for this comparison.

During the five years before 2004, the mover has seven Amkor patents in these classes, six of which cite the IBM patent, giving a citation rate of $6/7 = 85.7\%$. During the five years beginning in 2004, the mover has seven patents in the same classes assigned to the destination firm, none of which cites the IBM patent, so the citation rate falls to zero. The mover's citation rate therefore declines by 85.7 percentage points.

Three staying Amkor inventors satisfy the sample restrictions for this comparison. Their average citation rate is 57.6% before 2004 and 54.8% afterward, a decline of 2.8 percentage points. The difference-in-differences for this comparison is therefore $-85.7 - (-2.8) = -82.9$ percentage points.

The example also illustrates the random benchmark. There are four inventors in the comparison: the mover and the three stayers. Assigning the mover role to each of them in turn produces four possible pseudo-difference-in-differences: -82.9, 114.3, -53.3, and 21.9 percentage points. Their mean is zero by construction, and the actual difference-in-differences is the smallest of the four.

Main Results. Figure 4 reports the citation rates before and after the move. Before the move, movers and stayers have similar citation rates: 61.7% of the mover's patents in the relevant

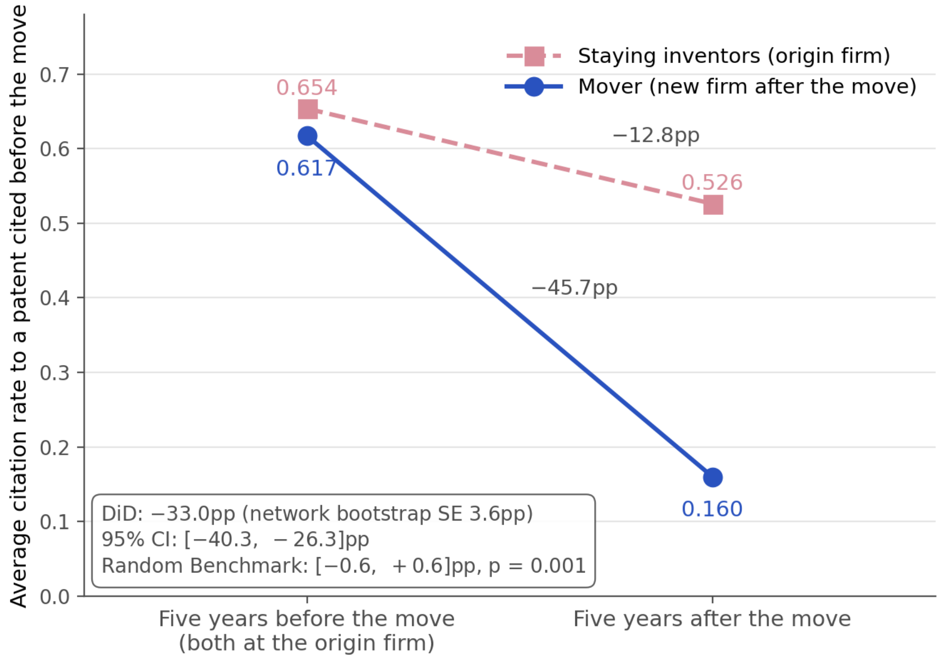


Figure 4: Citation Rates Before and After Inventor Moves

The figure compares citation rates for inventors who begin patenting at a new firm (movers) with those of comparable inventors who remain at the origin firm (stayers). Citation rates are the share of patents that cite a focal patent previously cited by both inventors. For movers, the figure compares patents filed at the origin firm during the five years before the move with patents filed at the destination firm during the five years after the move; for stayers, both periods use patents filed at the origin firm. In both periods, I restrict attention to the narrow technological classes in which the mover had cited the focal patent before the move. For each mover, I average the citation rates of the eligible stayers. The difference-in-differences compares the mover's before–after change with that of the stayers. The 95% confidence interval is based on 1,000 bootstrap draws clustered by connected components of the mover–stayer inventor network. The Random Benchmark shows the 2.5th–97.5th percentile range from 1,000 draws that randomly reassign the mover identity within each comparison.

technological classes cite patent q, compared with 65.4% for stayers. After the mover begins patenting at the destination firm, the mover's citation rate falls to 16.0%, while the citation rate among stayers falls to 52.6%. Thus, the mover's citation rate declines by 45.7 percentage points, compared with 12.8 percentage points among stayers, yielding a difference-in-differences of -33.0 percentage points. The network-clustered bootstrap standard error is 3.6 percentage points, with a 95% confidence interval of $[-40.3, -26.3]$. The result is also far outside the random benchmark. Its 2.5th–97.5th percentile range is $[-0.6, 0.6]$ percentage points, and the finite-randomization left-tail p-value is 0.001. These results indicate that inventors become substantially less likely to cite previously used knowledge after they begin patenting at a new firm, relative to inventors who remain at the origin firm.

Additional Results: Attorneys. Citations may reflect the practices of the attorneys working on the patents. To separate the roles of firms and attorneys, I consider attorneys who work for multiple firms. For each such attorney and cited patent, I identify the first firm for which the attorney appears on at least five patents citing that patent. I then compare the attorney's later patents for this firm with the attorney's patents for other firms, restricting the comparison to patents in the same application year and narrow technological class.[13]

Table B3 in Appendix B shows that the citation rate is 31.0% for the established firm and 12.8% for other firms, a difference of −18.1 percentage points. The 95% confidence interval is $[-22.0, -14.4]$ percentage points, while the Random Benchmark is $[-0.2, 0.3]$. Thus, the persistence of citations is specific to the firm and cannot be explained by attorney-specific citation practices alone.

Robustness. Table B4 in Appendix B reports several robustness exercises. First, I vary the weighting in the main statistic. In the main analysis, I assign equal weight to each cited-patent–mover–destination-firm triple. Thus, movers and cited patents appearing in more triples have a larger influence on the average. I therefore reweight the observations in two ways: (i) I give equal total weight to each mover, and (ii) I give give equal total weight to each cited patent.

Second, I vary the sample and control group. I restrict the sample to moves through 2008 (rather than 2010), giving post-move patents more time to be granted before the patent data end. I also lower the requirement that movers and stayers have at least five pre-move patents citing the cited patent to three. I group stayers connected through co-invention and give equal weight to each group, so that a research team with several eligible stayers does not count multiple times in the control average. Finally, I restrict the control group to stayers located within 25 kilometers of the mover before the move, making movers and stayers more comparable in their local inventive environment.

Third, I group patents connected through parent–child application links into patent families and count each family once. This robustness addresses the possibility that post-move citations of stayers are driven by continuation patents before the move.

In all these specifications, the difference-in-differences remains negative and statistically significant.

Finally, I analyze whether movers and stayers had diverging citation patterns before the move. Figure B5 shows average citation rates during the two five-year periods before the move and the five-year period after the move. The difference-in-differences between the two pre-move periods is small and statistically insignificant, while the post-move difference-in-differences is negative and statistically significant.

[13] The main difference from the inventor-movement analysis is that I compare the same attorney across firms in the same year rather than construct a group of "staying attorneys." An attorney appearing on patents assigned to multiple firms in the same year can naturally be interpreted as representing multiple clients. For inventors, appearing on patents assigned to multiple firms in the same year is harder to interpret because patent assignment is typically related to the inventor's employment at the assignee.

4 Rising Technological Complexity

This section develops a model that explains the increasingly concentrated and clustered citation patterns documented in Sections 3.2 and 3.3. I then provide evidence consistent with the model's predictions.

4.1 Model

The model shows how technological complexity affects knowledge transfers across firms. As applications require more complementary pieces of knowledge, transfers become more selective but involve more of the firm's knowledge.

Environment. Suppose a firm owns two patents, A and B, each embodying a separate piece of knowledge. There is a unit mass of potential applications. An A-application requires only the knowledge embodied in patent A, a B-application requires only the knowledge embodied in patent B, and an AB-application requires both pieces of knowledge. I refer to A- and B-applications as *discrete* and to AB-applications as *complex*.

The share of complex applications is $\mu \in [0,1]$. The remaining applications are divided equally between the two discrete types, so each has share $(1-\mu)/2$. Each application is associated with a separate adopter and generates value v, independently drawn from a uniform distribution on $[0,1]$. To realize this value, the adopter must license the required patents from the firm: a discrete application requires one patent, and a complex application requires both.

Transferring knowledge A or B is costless. For discrete applications, an adopter can learn the relevant knowledge from the patent document and needs only a license for the right to use it. A complex application additionally requires *tacit* knowledge about how to combine A and B. The firm can provide this tacit knowledge by sending engineers to the adopter (Teece, 1977; Arora, 1995; Arora, Fosfuri, and Gambardella, 2001). In this case, the firm incurs a cost $c \in (0,1)$.

Consider a relationship between the firm and an adopter. The timing is as follows. First, both observe the application's type and value. Second, the firm makes a take-it-or-leave-it offer for the required patent license. Third, the adopter accepts or rejects. If it rejects, the application is not implemented. If it accepts, the firm grants the license and, for a complex application, transfers the tacit knowledge at cost c. Whenever indifferent, the adopter accepts.

If the application is implemented, the adopter files for a patent and cites the patents whose knowledge it uses. A discrete application therefore cites either patent A or B, while a complex application cites both.

Equilibrium. In a subgame-perfect equilibrium, the firm charges v for a discrete application, and the adopter accepts. For a complex application, trade occurs if and only if $v \geqslant c$. When trade occurs, the firm charges v and transfers the tacit knowledge at cost c.

This equilibrium implies the following citation patterns. A share $\frac{1-\mu}{2}$ of adopters cites patent A only, a share $\frac{1-\mu}{2}$ cites patent B only, and a share $\mu(1-c)$ cites both A and B.

I define the *breadth* of knowledge flows as the mass of adopters that use the firm's knowledge:

$$B(\mu) = \frac{1-\mu}{2} + \frac{1-\mu}{2} + \mu(1-c) = 1 - \mu c. \tag{4.1}$$

I define the *depth* of knowledge flows as the average number of pieces of the firm's knowledge used by an adopter, conditional on knowledge transfer:

$$D(\mu) = \frac{1 \cdot \frac{1-\mu}{2} + 1 \cdot \frac{1-\mu}{2} + 2 \cdot \mu(1-c)}{B(\mu)} = 1 + \frac{\mu(1-c)}{1-\mu c}. \tag{4.2}$$

The following result follows from these definitions:

Proposition 1. *As the share of complex applications μ increases, knowledge flows become narrower but deeper: $B'(\mu) < 0$ and $D'(\mu) > 0$.*

The intuition is a follows. Transferring knowledge for a complex application is costly. Thus, the transfer occurs only when the application is sufficiently valuable. As the share of complex applications increases, the firm transfers knowledge to fewer adopters. But conditional on transfer, complex applications use multiple pieces of the firm's knowledge, increasing the number of knowledge pieces transferred within each relationship.

The proposition maps directly to citation patterns. As the share of complex applications increases, the firm's patents receive citations from fewer adopters, while those adopters increasingly cite both patents together.

Extension. I use the setup with two patents for simplicity. More generally, applications may require n pieces of knowledge embodied in n patents. Let c_n denote the cost of transferring the tacit knowledge needed to combine them, with c_n increasing in n. An application requiring n patents is then implemented only if $v \geqslant c_n$. Thus, applications requiring more patents are less likely to receive the firm's knowledge, but conditional on transfer, they use more of its patents.

4.2 Evidence

In this section, I provide evidence consistent with Proposition 1. I first document an increase in the share of patents in complex technologies. I then show that complex technologies have narrower but deeper knowledge transfers.

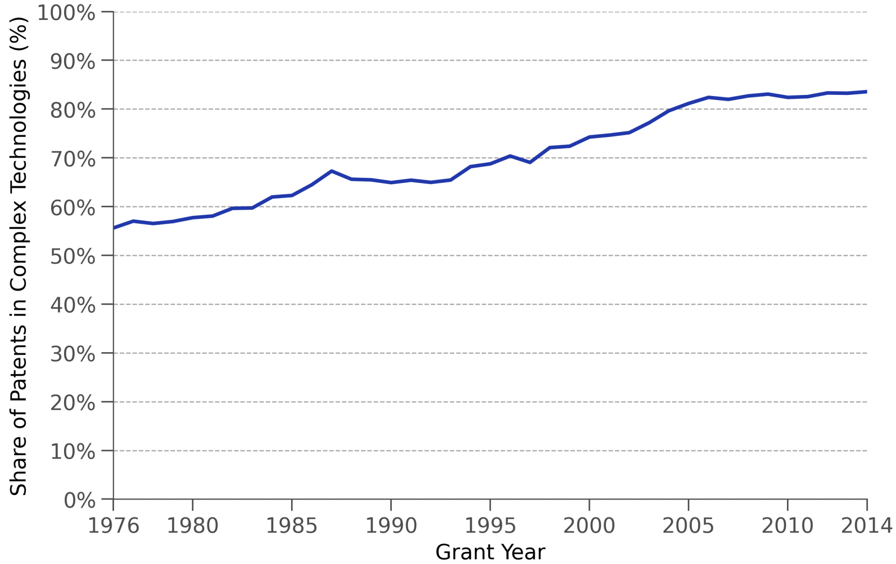


Figure 5: The Rise of Complex Technologies

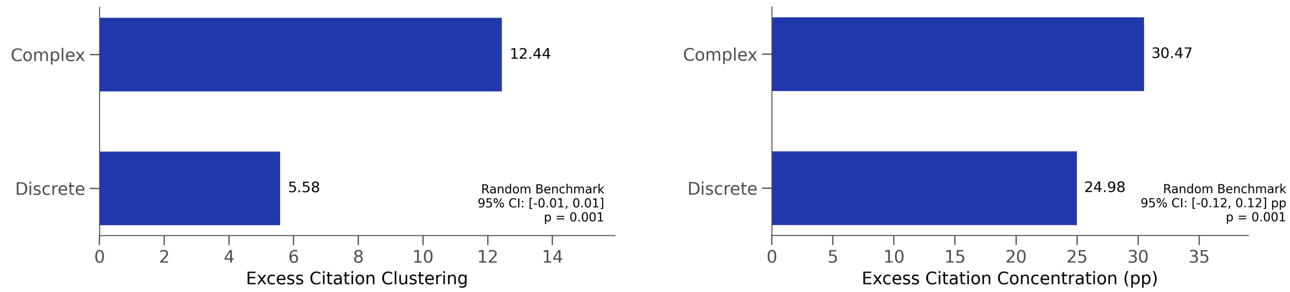
The figure shows, by grant year, the share of patents in technologies classified as complex. The sample includes single-assignee corporate patents granted between 1976 and 2014. Technological fields are classified as complex or discrete using the WIPO classification.

I classify technological classes as *discrete* or *complex* using the WIPO technology classification. Discrete technologies, such as pharmaceuticals, combine relatively few patentable components, while complex technologies, such as semiconductors, combine many. I use the share of patents in complex technologies as an empirical proxy for μ, the share of complex applications in the model. I consider patents assigned to a single firm and granted between 1976 and 2014.

Figure 5 shows the share of patents granted each year in complex technologies. This share increased from 55.6% in 1976 to 83.6% in 2014, indicating a substantial shift toward technologies that combine more patentable components.

I next compare discrete and complex technologies in terms of the breadth and depth of their knowledge flows, which in the model are defined in equations (4.1) and (4.2). I use the sample of top-cited patents from Sections 3.2 and 3.3.

I measure breadth of knowledge flows inversely by the concentration of citations across citing firms. For each cited patent, I separately compute the *excess* concentration of citations coming from complex and discrete technologies. The excess concentration is defined as the difference between the actual concentration and the random benchmark that preserves patent counts (see the definition in Section 3.2). It isolates the differences across firms in their citation rates. Thus, greater excess concentration corresponds to narrower knowledge flows and reflects larger differences in citation rates across potential citing firms.



(a) Citation Clustering

(b) Citation Concentration

Figure 6: Technological Complexity and Knowledge Flows

The figure compares complex and discrete technologies using the sample of top-cited patents from Sections 3.2 and 3.3. Panel (a) shows excess citation clustering: for each citing patent of a top-citing firm, I count the number of other patents assigned to the cited firm that it also cites and subtract the mean under the patent-count-preserving randomization described in Section 3.3. Panel (b) shows excess citation concentration: for each cited patent, I compute the concentration of citations across citing firms separately for complex and discrete citing technologies and subtract the corresponding mean under the patent-count-preserving randomization described in Section 3.2. In both panels, the "Random Benchmark" reports the 95% interval and p-value for the difference between complex and discrete technologies across 1,000 randomizations.

I measure depth of knowledge flows by the number of co-cited patents. As in Section 3.3, I consider patents of a top-citing firm that cite a highly cited patent and count how many other patents assigned to the cited firm they also cite. I define *excess* clustering as the difference between this number and the random benchmark that preserves patent counts. Excess clustering captures the extent to which a citing patent draws on additional knowledge of the cited firm beyond what the cited firm's patenting in the relevant technological areas would predict. Thus, greater excess clustering corresponds to the use of more pieces of knowledge from the cited firm.

Figure 6 shows that complex technologies exhibit both greater excess concentration and greater excess clustering than discrete technologies. Thus, in complex technologies, citation rates are more heterogeneous across potential citing firms, corresponding to more selective knowledge flows in the model. At the same time, citing patents in complex technologies cite more additional patents of the cited firm, corresponding to applications that rely on more pieces of the firm's knowledge.

5 Conclusion

This paper provides evidence that knowledge flows measured by patent citations likely reflect intentional knowledge transfers between firms. Over time, these transfers have become more selective across firms but more intensive within firm relationships. I develop and test a model in which rising technological complexity can explain these patterns.

I conclude by suggesting two avenues for future research. First, the model predicts a positive relationship between the complexity of applications and their average value conditional on implementation. Testing whether patents that rely on more pieces of knowledge from the same firm are more valuable is a natural direction for future research.

Second, this paper focuses on intentional knowledge transfers across firms. A firm's citations to its own patents can be interpreted as intentional knowledge transfers within firm boundaries. Comparing knowledge transfers within and across firms may therefore provide a way to test theories of firm boundaries and knowledge sharing (Teece, 1980; Grant, 1996; Liebeskind, 1996; Fadeev and Powell, 2026).

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A Data Details

Patents. The main source of patent data is PatentsView. Autor et al. (2020) provide a matching of patent assignees to Compustat firms for publicly traded companies. I use their matching for 1976–2014 and extend it through 2019. For the remaining patents, I follow the procedure outlined in Autor et al. (2020) to clean and standardize assignee names (e.g., replace "Incorporated" with "INC"). Finally, I manually match around 100 thousand patents belonging to the largest assignees.

FactSet Revere. I use two data sets from FactSet. FactSet Revere Company provides basic information on companies, including their names. FactSet Revere Supply Chain Relationships provides information on business relationships between firms. I discuss the relationship types below. The following quote from FactSet's manual describes how the data are collected:

> "FactSet analysts systematically collect companies' relationship information exclusively from primary public sources such as SEC 10-K annual filings, investor presentations, and press releases, and classify them through normalized relationship types. Company information is fully reviewed annually, and changes based on corporate actions are monitored daily. The result is a comprehensive, detailed and up-to-date dataset of material intercompany relationships."

FactSet also provides the names of subsidiaries involved in some business relationships. I match these names, including subsidiary names, to patent assignees using the same procedure as for the patent data.

FactSet records licensing relationships together with corresponding supplier–customer relationships. Specifically, a firm receiving a license is also recorded as a customer, while the firm providing the intellectual property is recorded as a supplier. For firm pairs with a licensing agreement, I therefore exclude the corresponding supplier–customer relationship.

Compustat Segments. I obtain supplier–customer relationships from Compustat Segments. For publicly traded firms, the data report the names of major customers, which are mostly firms but can also include government agencies. SFAS No. 131 requires publicly traded firms to disclose any customer accounting for more than 10% of total sales. Using Compustat Segments, Barrot and Sauvagnat (2016) construct a data set of supplier–customer relationships among U.S. publicly traded firms for 1976–2013. I extend their data through 2022 using name matching and manual inspection.

USPTO Patent Reassignment. The patent reassignment data are described in Graham, Marco, and Myers (2018). I retain only patent transfers between companies, formally restricting the data to transactions with 'convey_ty' equal to 'assignment'. I clean company names using the same procedure as for patent assignees and then match them to firms in the patent data.

All three data sets — FactSet, Segments, and Reassignment — provide information on the date of a transaction or a relationship. These dates are self-reported, so they might not correspond to the dates of actual relationships. For each firm-pair, I find the minimum and the maximum of years in which the relationship between firms was active. I leave only relationships active between 2003 (the first year in FactSet) and 2022.

I group certain relationships into more aggregated groups. Table B1 in Appendix B provides a summary of all partner relationships.

B Figures and Tables

Table B1: Types of Business Partnerships Between Firms

Group	FactSet Variables	Data	Share of Partner Pairs	
			Any in group	Equal split across groups
Suppliers and Customers	`SUPPLIER`, `CUSTOMER`, `PARTNER-MANUFAC`, `PARTNER-MARKTNG`, `PARTNER-DISTRIB`	FactSet, Compustat Segments	83.1%	68.9%
Research Collaboration	`PARTNER-RESCOLB`, `PARTNER-TECHNGY`	FactSet	29.7%	16.7%
Licensing	`PARTNER-LICENIN`, `PARTNER-LICENOT`, `PARTNER-PLICENS`	FactSet	13.0%	7.0%
Ownership Stakes	`PARTNER-EINVEST`, `PARTNER-INVESTO`, `PARTNER-JVENTUR`	FactSet	9.4%	5.3%
Joint Product Offering	`PARTNER-INTPROD`	FactSet	3.2%	1.2%
Unknown Relationship	`PARTNER-NA`	FactSet	1.7%	0.7%
Patent Reassignment	—	USPTO	0.4%	0.3%

Notes: This table lists the groups of business relationships used in Section 3.1 and the FactSet variables included in each group. Suppliers and customers are entities from which the source company purchases, or to which it sells, goods and services. Manufacturing, marketing, and distribution variables record the paid provision of these services and are classified as supplier–customer relationships. Licensing covers entities licensing products, patents, intellectual property, or technology to or from the source company. Research collaboration covers entities collaborating with the source company on the research and development of products not yet marketed. Ownership stakes cover entities owning equity in the source company, entities in which the source company owns equity, and jointly owned companies. Joint product offering covers entities with which the source company bundles otherwise standalone products or services into a single marketed offering. Unknown relationship denotes partners for which FactSet does not report a relationship type; these partners account for 0.7% of citations from partners. Patent reassignment, based on USPTO records, links firms that have transferred patents to one another. Descriptions follow FactSet's *Data and Methodology Guide* where an exact entry exists. The variables `PARTNER-TECHNGY` and `PARTNER-PLICENS` have no exact guide entry and are classified as research collaboration and licensing based on their variable names. The final two columns report shares of the 797,725 partner pairs of publicly listed U.S. firms. In the first, a pair counts toward every group in which it has at least one relationship. Because a pair may have relationships in multiple groups, these shares do not sum to one. In the second, each pair is divided equally across the groups in which it has relationships, so the shares sum to one.

Table B2: Share of Citations Between Business Partners: Main Result and Robustness

	Actual	Random Benchmark [2.5th–97.5th pct.]	Gap (pp)	Number of pairs
Front page, all citations	73.1%	21.0% [20.9, 21.1]	52.1	1,207,167
In-text citations	74.8%	18.9% [18.6, 19.3]	55.8	35,203
Front page: examiner-added	77.9%	23.2% [23.1, 23.3]	54.7	463,873
Front page: applicant-added	69.6%	20.4% [20.3, 20.5]	49.2	363,316
Front page, 10-year window	70.1%	20.0% [20.0, 20.1]	50.0	2,457,226
Front page, cited 2003–2014	73.4%	21.4% [21.3, 21.5]	51.9	1,015,695
Foreign public firms (front page)	71.2%	5.8% [5.7, 5.8]	65.5	590,939

Notes: This table reports the share of citations between business partners across different samples. The first row corresponds to the main sample described in Section 3.1. The remaining rows report robustness checks using: (i) in-text citations from Verluise et al. (2026); (ii) examiner-added citations; (iii) applicant-added citations; (iv) a ten-year citation window; (v) cited patents granted from 2003 onward, when FactSet coverage begins; and (vi) publicly listed foreign firms, defined as Compustat North America and Compustat Global firms headquartered outside the United States. The table reports the actual partner share, the partner share under the random allocation of citations described in Section 3.1, the difference between the two, and the number of patent pairs in each sample.

Table B3: Firm Specificity of Attorney Citations

	Established firm	Other firms	Difference (pp)	SE (pp)	95% CI (pp)	Random Benchmark [2.5th–97.5th pct.]
Citation rate	31.0%	12.8%	−18.1	1.9	[−22.0, −14.4]	[−0.2, 0.3]

Notes: This table compares citation rates across firms for the same attorney. For each attorney and cited patent, I define the established firm as the first firm for which the attorney appears on at least five patents citing that patent. I then compare the attorney's later patents for the established firm with patents for other firms in the same CPC main subgroup and application year. The sample contains 87,665 established attorney–firm–cited-patent relationships involving 1,507 attorneys. The difference is the citation rate for other firms minus that for the established firm. Standard errors and 95% confidence intervals come from 1,000 bootstrap draws clustered by attorney. In the Random Benchmark, I reshuffle the established-firm and other-firm labels within each attorney–class–year comparison, keeping fixed the number of patents and citations. The table reports the 2.5th–97.5th percentile range across 1,000 draws; the left-tail p-value is 0.001.

Table B4: Inventor Moves and Citation Rates: Main Result and Robustness

	DiD (pp)	SE (pp)	95% CI (pp)	Random Benchmark [2.5th–97.5th pct.]
Main	−33.0	3.6	[−40.3, −26.3]	[−0.6, 0.6]
Equal weight: mover	−28.9	2.1	[−33.1, −25.0]	[−2.0, 2.0]
Equal weight: focal cited patent	−33.7	3.4	[−41.0, −27.4]	[−0.8, 0.9]
Move year $\leqslant$ 2008	−30.7	5.3	[−41.3, −20.7]	[−0.9, 0.9]
Lab-component-weighted stayers	−33.0	3.6	[−40.2, −26.5]	[−0.7, 0.6]
Stayers within 25 km	−23.9	4.5	[−33.2, −16.0]	[−0.8, 0.9]
Patent families	−14.3	6.7	[−27.4, −2.1]	[−1.1, 1.0]
At least 3 pre-move citing patents	−30.0	3.4	[−36.3, −22.8]	[−0.4, 0.4]

Notes: This table reports the difference-in-differences in citation rates for inventors who begin patenting at a new firm (movers) relative to comparable inventors who remain at the origin firm. The first row corresponds to the main specification described in Section 3.4. The remaining rows report robustness checks using: (i) equal total weight for each mover; (ii) equal total weight for each cited patent; (iii) moves through 2008 rather than 2010; (iv) groups of staying inventors connected through co-invention, with citation rates first averaged within each group and then equal weight given to each group; (v) staying inventors located within 25 km of the mover before the move; (vi) patent families rather than individual patents; and (vii) at least three, rather than five, pre-move patents citing the same patent. The table reports the difference-in-differences, its standard error and 95% confidence interval, and the Random Benchmark. To calculate standard errors, I connect each mover to the staying inventors used as controls and group together all inventors connected through this network. I then resample these groups in 1,000 bootstrap draws. For the Random Benchmark, I randomly assign the mover role among the mover and staying inventors within each comparison and repeat the procedure 1,000 times. The table reports the 2.5th–97.5th percentile range across these draws and the corresponding left-tail p-value.

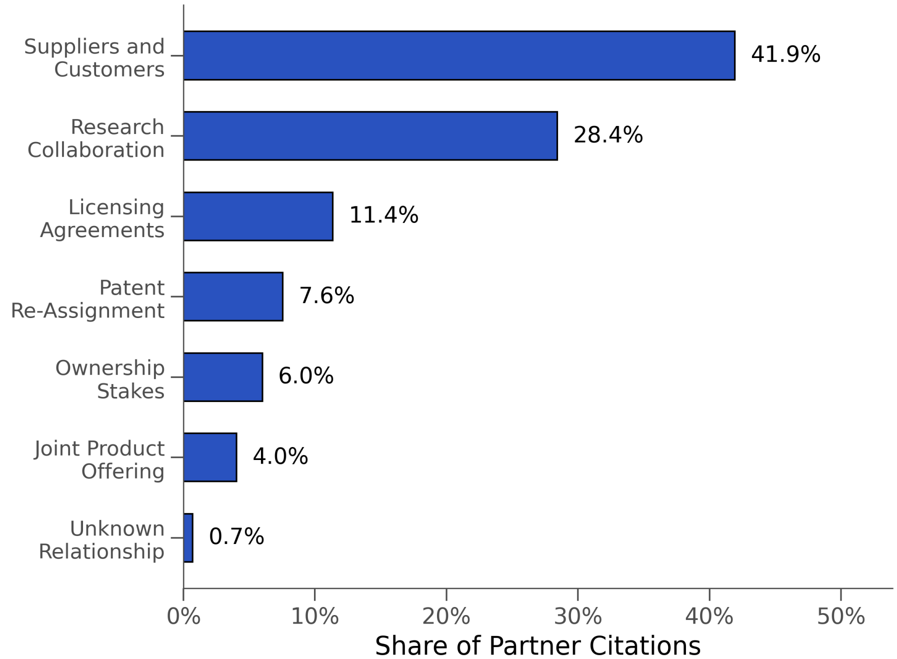


Figure B1: Distribution of citations from business partners by type of relationship

This figure shows the distribution of citations among business partners across different types of partnerships. The sample is the main sample described in Section 3.1. When two firms have more than one type of business relationship, I divide each citation equally across the relationship types.

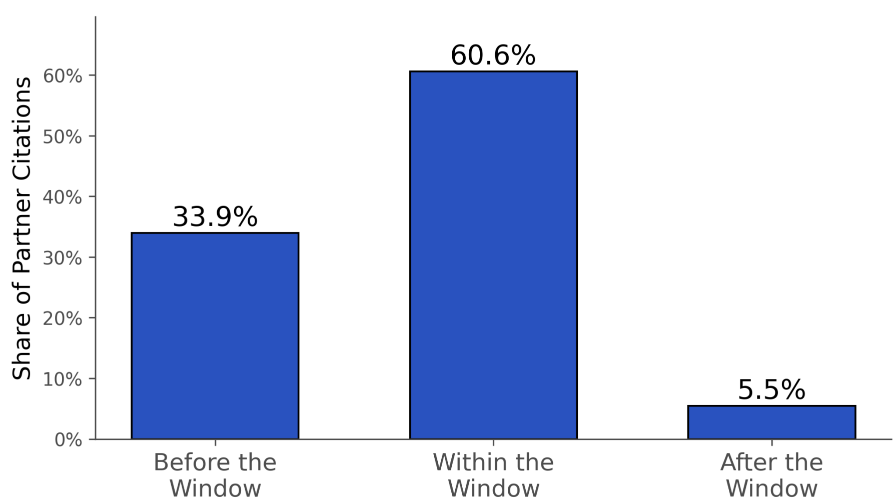
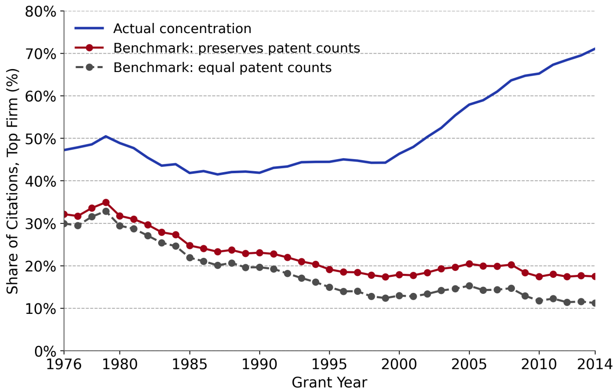
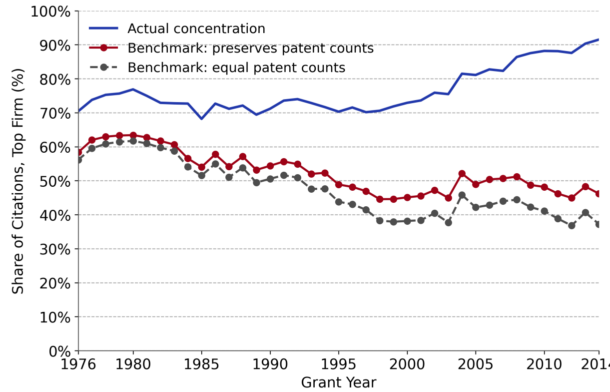


Figure B2: Timing of partner citations relative to the reported window of the partnership

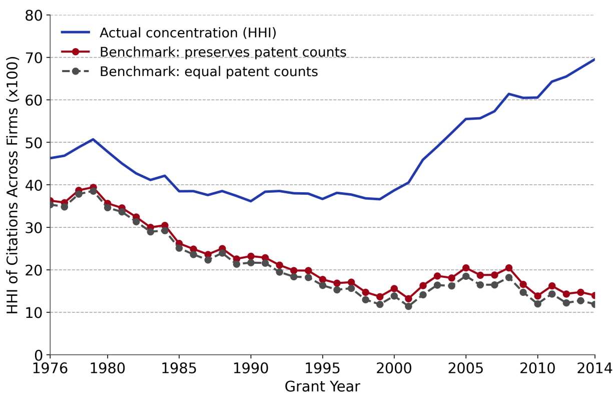
The figure shows the share of partner citations whose citing patent was applied for before, during, or after the reported partnership window. For partnerships reported as ongoing, I treat the window as beginning in the reported start year and continuing thereafter.



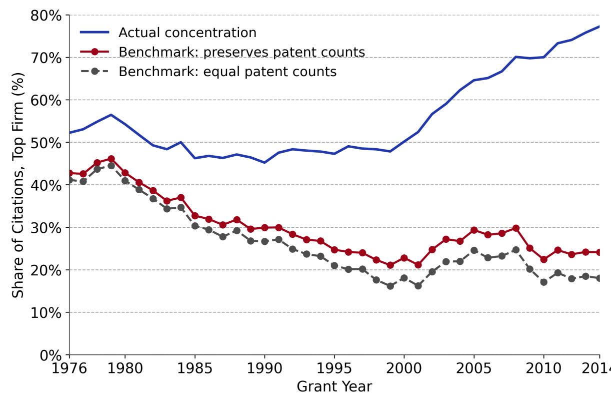
(a) Top 5% of the most-cited patents



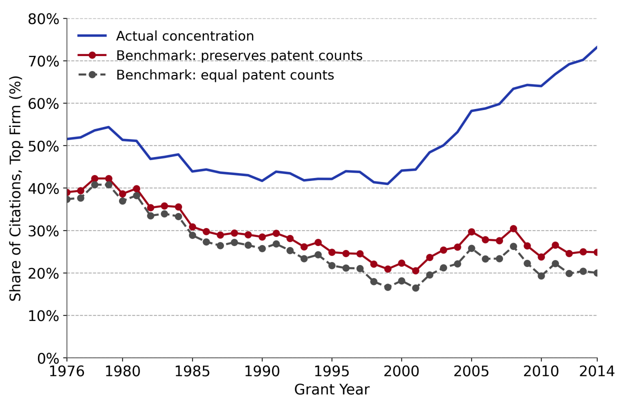
(b) Publicly listed U.S. firms



(c) Herfindahl index of citation shares



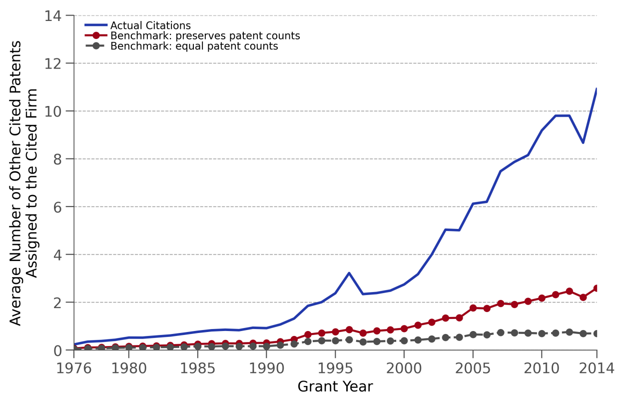
(d) Textual-similarity benchmark



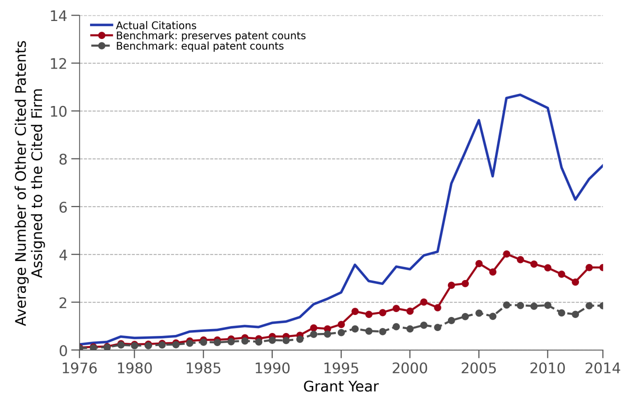
(e) Citing patents collapsed to families

Figure B3: Concentration of Citations: Robustness

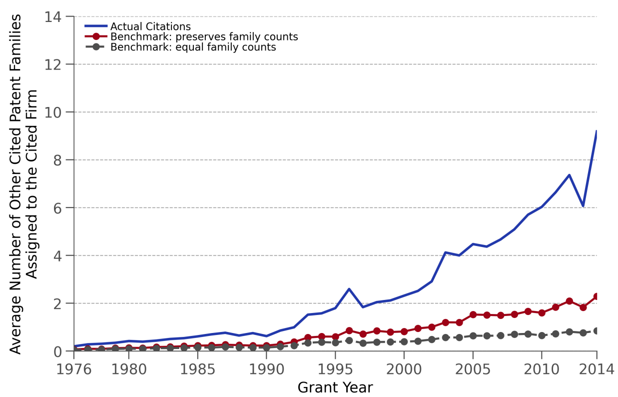
Each panel repeats the construction of Figure 2. Panel (a) restricts the sample to the top 5% of the most-cited patents within each grant year and technological class; panel (b) to patents assigned to publicly listed U.S. firms, as in Section 3.1. Panel (c) measures concentration by the Herfindahl index of citation shares across citing firms instead of the top-firm share. In panel (d), the random benchmark is computed on groups restricted by textual similarity: a non-citing patent remains in a group only if its PaECTER similarity to the citing patents is at least as high as the minimum pairwise similarity among the citing patents; the actual concentration is identical to Figure 2 by construction. In panel (e), citing patents are collapsed to patent families with one citing patent per family, represented by its earliest member. Benchmark lines report means across 1,000 draws.



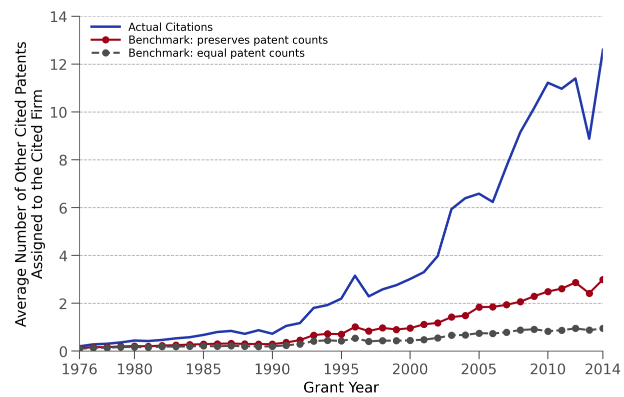
(a) Top 5% of the most-cited patents



(b) Publicly listed U.S. firms



(c) Cited patents collapsed to families



(d) Textual-similarity benchmark

Figure B4: Clustering of Citations: Robustness

Each panel repeats the construction of Figure 3. Panel (a) uses the top 5% of the most-cited patents within each grant year and technological class instead of the top 1%; panel (b) restricts the sample to patents assigned to publicly listed U.S. firms, as in Section 3.1. In panel (c), patents receiving backward citations are collapsed to patent families, represented by their earliest member, and the clustering measure counts distinct families rather than distinct patents. The randomization pool is collapsed to families in the same way. In panel (d), the randomization pools are additionally restricted by textual similarity. A non-cited patent remains eligible only if its PaECTER similarity to at least one actually cited patent is at least as high as the minimum pairwise similarity among the actually cited patents. The actual clustering measure in panel (d) is identical to Figure 3 by construction. Benchmark lines report means across 1,000 draws.

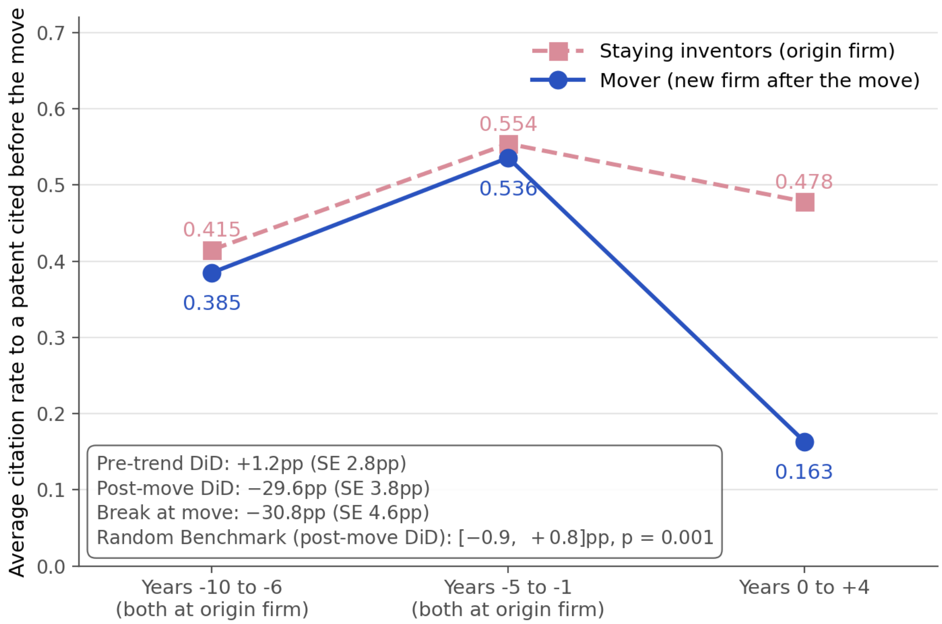


Figure B5: Citation Rates Before and After Inventor Moves: Pre-Trends

The figure compares citation rates for movers and staying inventors over three five-year periods. The first two periods cover years 10–6 and 5–1 before the mover begins patenting at a new firm; in both periods, patents of movers and stayers are assigned to the origin firm. The final period covers the first five years after the move, using patents assigned to the destination firm for movers and to the origin firm for stayers. In all periods, citation rates are computed within the narrow technological classes in which the mover had cited the focal patent before the move. The balanced sample contains 13,620 cited-patent–mover–destination-firm triples for which the mover and the included staying inventors have patents in all three periods. The pre-trend DiD compares the change across the two pre-move periods; the post-move DiD compares the second pre-move period with the post-move period; and the break at the move is the difference between these two DiDs. Standard errors are based on 1,000 bootstrap draws clustered by connected components of the mover–stayer inventor network.